William O. Holden

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Summary

AI Engineer building internal agentic and retrieval-augmented platforms at Aderant — three applications in seven months, each automating manual work for the teams behind the company's legal software. Background spans full-stack development, LLM orchestration, and deep learning for computer vision and NLP, with peer-reviewed research in *Computers and Electronics in Agriculture*.

Professional Experience

Aderant **Dec 2025 – Present**
AI Engineer, Docketing & VI Engineering *Atlanta, GA (Hybrid)*

- Designed and delivered **three internal AI applications in seven months**, each replacing a manual process for the teams that maintain Aderant's docketing and legal-content products.
- Built an **internal change-detection platform** that monitors **60,000+ court, agency, and judicial forms** in the Forms Workflow content library, surfacing revisions to the content team as structured, reviewable diffs instead of manual periodic checks.
- Architected an **agentic court rules monitoring tool** for Aderant's internal rules-monitoring team: crawls court websites across U.S. jurisdictions via **Browserbase** remote browsers, filters DOM noise to isolate true-positive changes, and applies **Claude** reasoning via Bedrock to judge whether a change materially affects a rule or filing deadline — sharply reducing false-positive review volume.
- Developed an internal **competitive intelligence hub** for product, marketing, and sales: deduplicates external news signals alongside **Aha!** and **Marketo** data into a **RAG** pipeline over a vector store, letting teams chat with the corpus and auto-generate battlecards, newsletters, and briefs grounded in cited sources.
- Embedded an **SDLC agent across the team's internal tooling** that intercepts inbound bug reports and enhancement requests, validates and normalizes them with an LLM, then routes clean, deduplicated tickets straight into **GitHub Projects** and Slack, removing manual triage from intake.
- Stack: Python, TypeScript/React, **AWS** (Bedrock, AgentCore, Lambda, ECS/EKS, EC2, Aurora/RDS, S3, Route 53), **Claude Platform**, Browserbase, Terraform, Flux GitOps, Grafana; built with Claude Code and Cursor.

University of Georgia — School of Computing **2024 – Dec 2025**
Graduate Research Assistant, Computer Vision *Athens, GA*

- Built a spatiotemporal video encoder with metadata fusion (X3D 3D-CNN + YOLO tracking) for automated broiler-chicken behavior recognition, trained on 200+ hours of smartphone farm footage.
- Developed motion-correction algorithms to normalize avian kinematics and remove human-handling bias, enabling reliable velocity and acceleration metrics from handheld video.
- **1st Place**, UGA Artificial Intelligence in Livestock and Poultry Farming Conference (poster).

Selected Projects

DeckTection — Trading Card Identification Pipeline *PyTorch, YOLOv12, SAM2*

- Engineered an end-to-end CV pipeline (YOLOv12 → SAM2 → Siamese network); fine-tuned YOLOv12 to **99.49% mAP50** and trained a ResNet-18 Siamese network to **95% one-shot accuracy** across 4,000 cards.

Whisper Inference Optimization *ONNX Runtime, Quantization*

- Quantized the Whisper encoder to INT8, cutting model size **70%** with no material word-accuracy loss (LibriSpeech); implemented Winograd convolution for 1D layers for **60%** compression enabling mobile deployment.

Cinema E-Booking Platform — Full-stack booking system; led Agile sprints for a 5-engineer team *React, Django, SQL*

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, C++, C, SQL, R

AI & ML: Agentic workflows & LLM orchestration, RAG, vector search, prompt engineering, Claude Platform, Amazon Bedrock, AgentCore, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, SAM, ONNX

Cloud & DevOps: AWS (EC2, ECS, EKS, Lambda, S3, Route 53, Aurora/RDS, Bedrock), Terraform, Flux GitOps, Docker, Grafana, Git/GitHub Actions

Data & Web: React, Node.js, Django, REST APIs, Aurora/PostgreSQL, MongoDB, SQL Server, Aha!, Marketo

Tools: Claude Code, Cursor, Glean, Browserbase, Playwright, Postman

Education

University of Georgia **Athens, GA**
M.S. in Artificial Intelligence — GPA 3.73 *Dec 2025*

University of Georgia **Athens, GA**
B.S. in Computer Science, Honors Program — Presidential Scholar, Dean's List *Dec 2024*

Publication

Holden, W.O., Li, G., Agyemang-Duah, E., et al. "Spatiotemporal video encoder integrated with metadata fusion for action recognition and behavior analysis." *Computers and Electronics in Agriculture*, 2026 (IF 10.3).